Spatialisation, Method and Madness
Learning from Commercial Systems

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Abstract

Computer music systems have supported dynamic speaker map models for sometime. With the advent of multiple surround delivery formats, commercial film and TV mixing systems now also support automated down-mixing, albeit for limited range of speaker formats. This paper compares the methods and approaches to producing multi-channel material for multi-speaker replay in the various contexts of music, film and TV, theatre, and installation. The author's experiments in automated spatialisation developed in the PD composition environment are also introduced. In addition to this technical review, the paper also considers aspects of aesthetic and psychological significance in the presentation of sound material in multi-speaker replay systems.

1 Introduction (Method and Madness)

This paper provides an overview of the areas of interest that have influenced my experiments in developing an interface for multi-speaker diffusion. In some ways the process is more interesting than the outcomes, as my experiments are somewhat idiosyncratic and not generally applicable.

Recently, I have been working in a commercial environment as part of a team developing a new mixing system for post-production. This experience has opened my eyes to the priorities that influence the development of such systems and the demands of those professionals working with this style of equipment. These priorities represent a strange amalgam of method and madness, excellence and compromise. However, I have become aware that the technical infrastructure behind much contemporary music is subject to very similar forces.

1.1 Philosophy of Design

There are as many approaches to the design of multi-speaker sound reproduction systems as there are applications for presenting sound in space. It is also obvious that the creative applications to which technology is applied often subvert the original intentions of the designers.

Reproducing Reality Reproducing acoustic reality or synthesising a reasonable representation or reproduction of an acoustic event is the principle behind many systems. The most ubiquitous example of this approach is the theoretical foundation of the stereo reproduction system (Rumsey 1998). Other systems, which apply this approach, are: ambisonics (Malham 1990, 2000; Gerzon 1984, 1985, 1992); virtual reality and industrial simulation systems employing various approaches including very large speaker arrays and other methods of wavefront synthesis (Boone, et al 1996).

These types of systems attempt to recreate a sound field, or at least a part of the sound field.

Pragmatic Approaches Many, more pragmatic approaches attempt merely to present enough auditory information to give the listener the impression of sounds in spatial relation to each other. These systems often rely on intensity panning and the resultant limited time domain cues available within a limited listening area. Examples of these approaches included cinema surround sound systems and their associated encoding and panning methods (Stevenson 2001, 2002); vector based amplitude panning systems and their employment in musical systems and various entertainment installations (Pulkki 1997; Drost and Rolfe 1999)[1]; and triangulated panning methods (LCS 2002).

[1] Note: Chris Rolfe of Thirdmonk Software has indicated that the methods employed in ABControl

Spatial Encoding Spatial encoding methods also show significant diversity. Whereas A and B format ambisonic signals (Gerzon 1985) encode an entire sound field in a finite number of matrixed audio channels, other systems store automation or metadata along with a number single channel audio signals or pre-mixed mutichannel stems.

In many cases, these methods of encoding provide, to a greater or lesser degree, a level of speaker system independence. This allows a system that records positional information along side raw audio signals, to be decoded for playback on a speaker system of different proportions to the one on which the material was encoded. This feature usually requires the speaker system to conform to a basic set of design criteria specific to each encoding method. For example, a system may only contain dimensional information for a single plane, or may require the loudspeakers only to be positioned only on the perimeter of the listening area.

Some encoding systems allow spatial transformations to be performed on the encoded material. These processes include divergence, rotation, and soundfield contraction or expansion (Wishart 1996). Most commercial systems offer hybrid approaches, allowing the use of discrete, encoded and pre-mixed elements to be combined.

2 Overview of Systems

There are a wide range of acoustic and psycho-acoustic properties that may be employed and exploited, or ignored in the design of a multi-speaker replay system. In this paper I will merely list the significant parameters to give an idea of what has and hasn't been included in the various approaches. Please refer to the relevant literature for more detailed information.

The auditory system can detect a number cues which assist it in evaluating the properties of individual sound sources, their size, location and motion; properties of the environment, its material composition and disposition; and properties of the composite soundfield which is the result of the temporal, spatial, and timbral or spectro-morphological relations between individual sound sources, and the spatial relations between the sources and elements of the environment. Nearly every acoustic and psycho-acoustic parameter is interdependent and the designer must select a subset, or group the combination of effects to arrive at a set of parameters which can be controlled to produced a reasonable simulation of the spatial location of static and moving sound sources.

Of course, we do not only perceive sound as the result of bodies behaving according to the laws of physics. Sounds come to us in moments of reverie and confusion, in dreams, and from electronic music and the ever-present electronic media, which create synthetic sound images for which evolutionary psychology has not prepared us.

2.1 Potential System Parameters

Acoustic/Physical Properties The acoustical properties that may be considered in the specification of a diffusion system include:

- Source position – this is the primary physical property we are aiming to control. The perceived source position is to be determined relative to the listener and relative to the loudspeakers. We may need to consider that there is more than one listener and that the source and or the listener may be moving.
- Listener position – for large audiences this may be difficult to specify. However a listening area is usually assumed as part of the system specification. Lack of a single listener influences the range of parameters which will have effect. The listener position maybe used as a higher order control parameter in spatial transformations performed on an entire sound field.
- Relative intensity – although sound intensity diminishes in proportion to distance from the listener, intensity must be used in combination with other parameters to achieve an effective distance cue. The relative intensity of signals produced by each loudspeaker is the primary tool used to simulate the location of sound sources.

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software should be distinguished from the methods described by V.Pulkki. Informal tests have revealed significant differences in the performance of the two systems.

- Relative loudness – the relative loudness of two sources gives the listener information on the physical properties of sound sources. This includes the size and kinetic energy of the sources.
- Fixed propagation velocity of sound – the fixed propagation velocity of sound in air or any other stable medium results in a temporal offset between sounds from different source locations. Control of relative arrival time can be used in some situations but a range of listener positions may reduce its effectiveness. The fixed propagation velocity of sound also affects the perception of moving sources. This results in the Doppler effect where the pitch of moving sources changes with their angular velocity relative to the listener.
- Frequency dependent transmission properties of air and enclosed spaces - high frequency attenuation with distance in free air, influences the perception of relative spatial locations of sources, and the trajectory of moving sources. The frequency dependent transmission properties of enclosed spaces produces similar effects though not necessarily on high frequencies alone.
- Spectral balance – the overall spectral balance is related to various other physical properties.
- Reverberation time – the overall reverberation time is determined by the volume and material properties of an enclosed space.
- Direct/reverberant ratio – the ratio of direct energy to reverberant energy provides a strong cue to the distance of the source and the overall properties of the space.
- Direct/early reflection arrival time difference – the time delay between the arrival of the initial direct propagation signal and the first cluster of early reflections from nearby surfaces gives the listener cues to relative locations of sources and surfaces and the size and composition of space.
- Diffusion – the acoustical properties of diffusion relate to both the spatial distribution of reverberant energy and the coherence or correlation of that energy. These parameters effect both individual sources and the entire soundfield.
- Source frequency dependent directivity – the source directivity will effect the perception of position, distance, trajectory and orientation of the source.
- Interaction of interconnected spaces and objects within the environment – complex spaces will cause a range of acoustic interactions that may be modeled by a diffusion system.
- Occlusion and Exclusion – the transmission properties of the interface between spaces will effect the perception of spatial relations with the listening environment.
- Obstruction – transmission path obstructions will effect the perception of both static and moving sources. Reflection, refraction and diffraction effects all give clues as to the position and the physical properties of sources. These effects are often responsible for mistakes and confusion in the analysis of real acoustic environments.
- Acoustic properties of environmental materials – absorptivity and specularity affect the overall sound field.

Psycho-acoustic/Perceptual Properties These psycho-acoustic properties are responsible for creating the perceptual cues that our auditory system responds to. The psycho-acoustic properties that may be considered in the specification of a diffusion system include:

- Inter-aural time difference (ITD) – the difference in arrival time between the two ears is a strong cue for the position of a source in the horizontal plane. The auditory system has a higher resolution for sources lying in the front ninety degrees. ITD cues may be manipulated when two or more loudspeakers are used to reproduce a sound in space. The effectiveness of this approach is reduced

by inter-aural cross-talk and non-fixed listener positions.

- Inter-aural level difference (ILD) – otherwise known as inter-aural intensity difference (IID) is the primary means of creating the illusion of spatial source location. ILD and ITD interact through vector summation when continuous signals are panned between loudspeakers. However, the auditory system is not very effective in localizing continuous signals and continuous signal are not very common in music or the environment. Continuous signals are most likely to cause confusion in the perception of source location.
- Reverberation – the reverberation properties of moving sources as described above are strong cues for distance perception.
- Head related transfer function (HRTF) – the perception of source location in both horizontal and median plane is affected by the filtering properties of pinna, shoulders and head. Luckily this complex tool is built in to our hearing but its effects may be exaggerated in the replay system.
- Head movement – head movement is probably the most effective tool the auditory system has for discriminating source position. Head movement relies on all the other cues simultaneously. It also destroys or seriously modifies the most subtly created multi-speaker soundfield (however, this effect is reduced as the number of loudspeakers employed increases).
- Inter-aural correlation or diffusion – the similarity of signals presented to the two ears effects the evaluation of the acoustic properties of the environment.

I will briefly mention some of the systems and methods employed in various fields of sound production.

2.2 Computer Music Systems

Vector Based Amplitde Panning (VBAP) as described by Pulkki (1997) and as employed in Csound (http://www.csounds.com), Max/MSP (http://www.cycling74.com) and PD (http://www-crca.ucsd.edu/~msp/software.html) signal processing objects, describes the loudspeaker and source locations in terms of vectors. Signals are distributed between any three loudspeakers in a three-dimensional loudspeaker system. The signal level fed to each loudspeaker is modified by a gain coefficient determined by the length of orthogonal vector components. This method allows source locations and trajectories to be specified in terms of vectors in a unit sphere. Loudspeaker locations may be specified independently but must lie on the surface of the unit sphere. Total intensity is directly proportional to vector length therefore the vector representation does not reflect the actual source location and source trajectories may not pass through the centre of the space with contiguous vectors. As has been pointed out, the centre of the listening space is difficult to define for more than one listener.

Csound signal modifiers (Vercoe 1992) in addition to "VBAP" include "Pan", "Locsig", and "Space". These signal processing routines use intensity panning with the addition of sends to global reverberators. They allow the use of user definable pan laws but are limited to quad type loudspeaker configurations. Source trajectories may be read from a predefined table. Trajectories lie within the boundaries of the loudspeakers.

Symbolic Sound Kyma (Symbolic Sound Corp. 2001) includes a spatialisation system where source locations are specified in terms of polar coordinates relative to a unit circle. Source locations within the unit circle have a vector length of 0 – 1, outside the unit circle source locations approach the acoustic horizon at a vector length of 2. This system allows an intuitive relationship between source position and loudspeaker position. Trajectories may be entered on a graphical timeline alongside other signal processing functions including reverberation parameters. Source locations and speaker locations may be specified independently allowing for re-rendering spacialisations on different loudspeaker systems.

Gmebaphone/Acousmonium The Gmebaphone (Clozier 2001) and GRM Acousmonium are complex and unique systems employing intensity panning, reverberation, equalisation and loudspeaker design or

"registers" to modify the perception of individual sources and pre-mixed tracks. Loudspeakers are distributed throughout the auditorium. The unique acoustic properties of these systems are exploited in much the same way as a musical instrumental ensemble. It is my impression that adapting a piece for these ensembles would be considered similar to the process of re-orchestration of instrumental music

2.3 Virtual Reality, Auditory Display, Simulation and Research

This is a category of convenience that I am using to introduce spatialisation systems where spatial resolution is related to the number of loudspeakers employed. This is a general principle that may be employed in most systems. In auditory display systems, the human auditory system's increased ability to discriminate multiple signals through spatial separation is often exploited through the application of a loudspeaker-per-source approach. Wave field synthesis (Boone, et al 1996) and very large speaker arrays offer approaches to synthesising facsimiles of real sound fields.

2.4 Commercial Installation and Entertainment Systems

Level Control Systems SuperNova and CueStation (LCS 2002) products employ a method of defining a listening space or speaker map as a set of triangles or tri-sets of loudspeaker locations including virtual source locations. Virtual locations may distribute signals amongst other loudspeakers or may feed external signal processing systems for adding reverberation and smoothing trajectories. Virtual locations may also be placed beyond the perimeter of the listening area to provide an acoustic horizon. Signal intensity is modified in proportion to the distance from the source to any of the three vertices of the surrounding tri-set. This may result in the signal being sent to any number of loudspeakers. Source trajectories can be plotted on a graphical interface and the speaker map may be modified independently of the trajectories.

2.5 Other/Cross-over

This category includes systems that are employed in any of the above areas.

Richmond Sound Design AudioBox This hardware platform provides user defined pan law tables, a cross-point gain matrix, audio playback and a cue sequencing system. Any software capable of producing MIDI system exclusive messages can control the system. ABEdit (Clampett 2002) offers a rudimentary event based interface. ABControl (Drost and Rolfe 1999) offers a vector based panning system including algorithmic trajectory utilities. This vector based system offers loudspeaker independence but similar to other VBAP systems, the vector locations do not map directly to source locations within the unit circle.

Ambisonics (Gerzon 1984,1985,1992; Malham 1990,2001) as mentioned earlier, is a method of encoding an entire sound field. B Format ambisonic signals include three orthogonal velocity and one omni-directional pressure component. From these signals, discrete loudspeaker components can be derived for any position on a scaled unit sphere. Ambisonics provides a convenient means of specifying spatial information for discrete sources or recording acoustic events with specially design microphone arrays. In addition to its general applicability, ambisonic encoding allows high order spatial transformations, including rotation, and scaling, to be applied to the entire soundfield.

Ambisonics has been employed in multi-channel broadcasting, music recording, and computer music applications.

2.6 Film and TV Mixing

Multichannel audio formats have been a part of cinema since the 1930s. Dolby Surround, introduced in the 1970s made four-channel audio a standard in cinemas around the world. Dolby's mono and stereo compatibility enabled it to be introduced as a no additional cost added feature in stereo television broadcasting, requiring the consumer to invest in a consumer pro-logic decoder.

By the end of the 1980s digital audio had made its way into the cinemas and Dolby digital 5.1 become the standard. During the 1990s Sony's attempt at vertical market integration, following the purchase of CBS, saw the introduction of SDDS 7.1. More recently, the Dolby empire struck back with Dolby EX 6.1 for the most recent George Lucas "Star Wars" extravaganza. With the advent of DVD as a medium for domestic movie viewing and innovative music-in-

surround plus audio-visual content, media producers have had to contend with multiple delivery formats.

Luckily all these surround sound formats are two-dimensional and controlled by well adhered to standards. Cinema dubbing mix engineers have been very conservative in their use of surround effects. Due to extensive testing and trial showings mixers have learned what will and will not translate to different listening environments. Their priority has tended to be supporting the narrative and this translates to dialogue. Some practitioners such as the famed Walter Murch have managed to utilise spatialisation as a narrative tool. Interestingly, Murch cites Orson Wells mono radiophonic and cinema soundtracks as being his greatest influence (School of Sound Lecture, London 1997). Wells proved that not only could space be used as a narrative tool but that all the space you needed could be encoded onto a single low dynamic range, low bandwidth audio channel.

To briefly summarise the speaker layouts of the various different formats: mono is still the predominant format used by most televisions and radios; most formats must provide stereo compatibility for radio and television; Dolby Surround includes left, centre, right and a band limited surround channel. The surround systems in cinemas usually comprise an array of loudspeakers down each side of the cinema. Dolby Pro-logic 2 includes two surround channels, the domestic standard allows for two discrete surround loudspeakers; Dolby Digital 5.1 and DTS digital both provide five full range loudspeakers plus one sub-bass channel; Sony Dynamic Digital Sound (SDDS) provides left, left-centre, centre, right-centre, right, left-surround, right-surround, plus sub-bass (this arrangement of five loudspeakers across the screen was first introduced in 1954 for the 70mm wide-screen releases of the musicals “Oklahoma” and “South Pacific”); Dolby EX introduces a centre surround channel allowing the rear soundfield to be more consistent and integrated by filling the hole-in-the-middle where a phantom image may otherwise have existed.

Any mixing system intended for use with such a range of target formats must be inherently flexible in its mapping of spatial position to speaker system. Many current post-production systems allow material to be mixed in more than one format simultaneously with monitoring systems that can be switched between formats. Two such systems are the DSP Media Postation 2 (Stevenson 2001) and the Fairlight DREAM Console (Stevenson 2002). These products are integrated disk recorders editors and mixers. Both systems allow channel panning independent of speaker format. The Postation offers a touch-screen panning interface which may be superimposed on top of the video image. Pan position is recorded along side other mix automation data as a series of X,Y co-ordinates. When played back these coordinates are translated into individual bus element gain coefficients determined by their position relative to the loudspeakers in a virtual map of the soundfield.

Fairlight’s DREAM system offers several additional features. The monitoring system provides a user modifiable matrix of bus element down-mixing coefficients to support differences between mixing and monitoring requirements for domestic and theatrical target formats. These differences are primarily in the area of bass management. The DREAM system includes support for pre-mixed surround stems. Stem mixes are often produced separately for dialogue music and effects and may include many separately spatialised tracks. Multi-channel stems may be controlled like any other signal path with provision for level, equalisation, dynamic range control, grouping and routing. In addition to these features, higher order spatial transformations performed on an entire soundfield, such as rotation, contraction, expansion and divergence can be performed on a premixed stem. All these controls may be automated, trimmed and edited during the mixing process.

Panning control in the DREAM system is provided with motorised joysticks. This brings up the interesting problem that the spatial mapping system must satisfy the conflicting requirements to be both a useable physical control interface for specifying perceived source location in real-time plus an accurate space map for up or down-mixing to other formats. Computer music practitioners who have used joystick during diffusion performance will know that the equal power pan curves used in stereo panning systems do not translate well to the psychological expectation of joystick response. In fact a linear relationship of joystick position to source location is often more acceptable. Similarly the curves suitable for left to right panning are not necessarily appropriate for front to back panning. Not only is the auditory system less discriminating in this axis but

the speaker systems on cinema mixing stages have radical differences in type and topology between the screen and surround system. The earlier Fairlight "Fame 2" system attempted to overcome this by providing user configurable pan tables that could be specified independently for left to right and front to back panning operations. This also fulfilled the requirement for –3dB equal power panning for music applications and other laws such as –4.5 dB at centre for broadcast applications (Rumsey 1998).

3 Pan-O-Rama (not another half-baked spatialisation system)

Having taken into consideration all of the possible approaches outlined above. I have attempted to develop an idiosyncratic system which fulfils the requirements which are of most importance to me.

3.1 Compositional Tool

This system is a compositional tool for sonic art as opposed to a hybrid performance tool such as the Gmebaphone. Although simple trajectories can be recorded and controlled in real-time, more general human computer interaction problems have not been dealt with. Trajectories can be defined and edited on-screen with a pointing device.

3.2 Arbitrary Speaker Placement

Some argue, with reasonable force, that the seating arrangements in an auditorium have as much impact on the reception of the music as some of the more subtle spatial relations of which the music is composed. In addition to this consideration, loudspeakers should not be restricted to lying on the perimeter of the listening area. From these points you should be aware that I am not concerned with creating accurate representations of real soundfields. Although the orientation of the soundfield with respect to the listener may well be critical in conveying some of the meaning bearing gestures of spatial music. Simulation and approximations and effects are the concepts I am working with.

3.2 Pan Through Centre

Although the centre of the listening area may not be well defined in any loudspeaker arrangement, the fly over is one of the most effective gestures in cinematic spatial language. It can also be used with some subtlety, if sparingly, in more refined works of acousmatic music. The control interface should relate source position directly to perceived source location as accurately as possible given the deficiencies of multi-speaker intensity panning noted earlier. This requires the source target to be moved through the listening area and into the distance beyond the loudspeakers.

3.3 High Order Spatial Transforms

The user interface is not quite there yet but the system must support effects such as rotation and contraction.

3.4 Speaker System Independent

The panning system must support the recreation of spatial effects on systems with different numbers and positions of speakers. The speaker layout can be edited as simply as the trajectories.

3.5 Front End GUI and Scripting Output Independent of DSP

Software such as PD, running on the latest general-purpose personal computers are now providing signal-processing resources adequate for simple spatialisation. Sequencing data can be read from simple text files with interpolating look-up tables providing gain coefficients in realtime.

3.6 Flexible Pan Laws

Pan laws can be generated arbitrarily and saved as text files to be loaded into interpolating look-up tables. Floating-point indexes allow pan tables to be of any size.

3.7 Support for Position Dependent Reverberation and Decorrelation

The system produces a distance parameter and allows the control of reverberation through the use of position information and scripted control of reverb time and direct/reverberant energy ratios.

3.8 Triangulated Pan Map

Energy is distributed between any three loudspeakers. Virtual loudspeaker positions allow for smooth panning in complicated loudspeaker layouts. These ideas are borrowed from the Level Control Systems SuperNova and CueStation products. However, the geometry model may be unique. Figure 1 demonstrates the method of generating indices for the pan table look-ups. Although this looks

complicated, it results in equal loudness for any source position within the triangle.

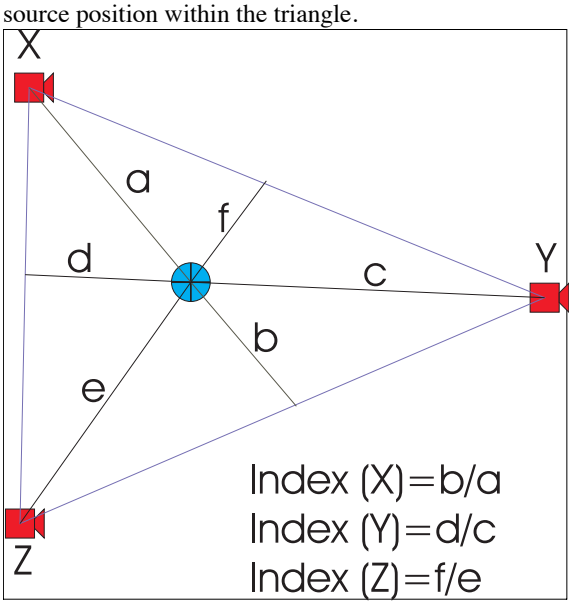


Figure 1. Triangulation Method

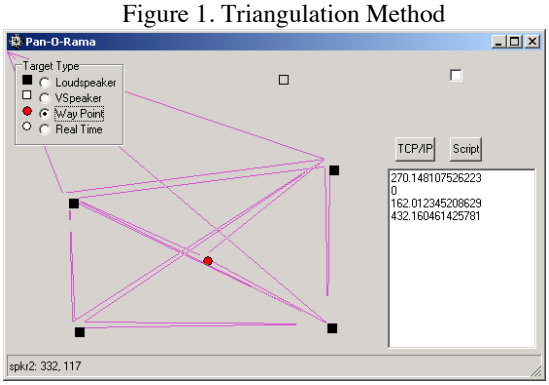


Figure 2. – The Pan-O-Rama Interface

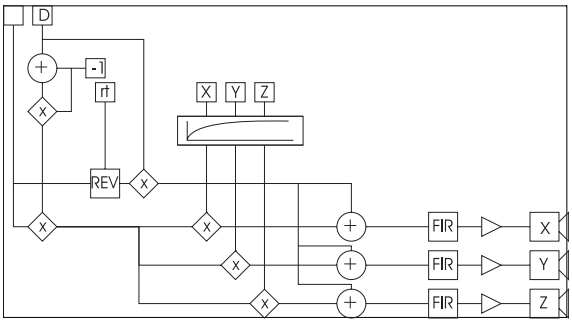


Figure 3. – Typical Signal Processing Arrangement

4 Aesthetic Considerations

Unfortunately, for a non-programmer, the work involved in developing projects such as this, tend to distract from the objective of making interesting work with sound. What, after all is the point of being able to whiz sounds around at will?

4.1 Validity of Space as Musical Property

Much debate surrounds the degree of importance to be attached to spatial manipulation as a significant element of musical language. The arena of musical practice is as broad today as it has ever been and there is undoubtedly continued scope for interesting work in this area. I would merely like to add some observations from areas not strictly aligned with contemporary music debate.

Elementary psychoacoustics tells us that the auditory system's capacity to distinguish discrete sonic events or streams is aided by spatial separation of sources. This explains part of the pleasure of listening to chamber music in the recital hall rather than at home on the record player. The other part of the explanation is the drinks at interval.

Cinematic language as I have mentioned earlier has occasionally managed to overcome the temptation for more meaningless special effects. Spatial effects in the soundtrack have been put to good use in supporting and augmenting the narrative path of drama. Additionally, surround sound can offer support to the less tangible aspects of developing non-verbal relationships between characters, locations, and spaces. This is the work that is otherwise done in the descriptive passages of literature which cinema must replace with sounds and images. Interestingly, the language of spatial cinema sound was developed primarily during an era when a monophonic audio track was the standard.

Illusion and Confusion in auditory perception is the area of most interest to me. Although I have done little research in this area I am convinced that there is something significant about the psychological effect of perceptually confusing sounds. In my experience, when the spatial origin of a sound is difficult to pin down, its cause or acoustic source is also difficult to determine. And the reverse is often true. This effect can be a strong stimulus to the imagination. You may call me a dreamer, but these states of imaginative reverie are exceptionally rewarding and often lead to a more "musical" perception of the natural soundscape. This process can often be triggered by the confusion caused by conflicting auditory cues when sounds a played back over multiple loudspeakers. In my case this either leads to intense frustration or enhanced absorption in the spectro-morphological content. I have to admit that too much

objective listening or reduced listening in the past has caused me to overlook these experiences until recent years when I have become less involved in other peoples music.

5 Conclusions

This wide ranging discussion has touched on the broad issues concerning the development of spatialisation systems for a diversity of applications. The somewhat idiosyncratic approach has been steered by my particular interests which I hope will serve as an introduction to the field for some and a stimulus for further debate for others.

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